

Yesun-Erdene (Yesuu) Jargalsaikhan

Portfolio | [Linkedin](#) | [GitHub](#) | she/her
Location: Trento, Italy

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EXPERIENCE

Research fellow, Developer, (full-time)

University of Trento

Apr 2026 – Now

Trento, Italy

- Role: Research in Natural Language Processing (NLP) with concept-based knowledge graphs; Development of multilingual lexico-semantic resources and web applications
- Reports to: Fausto Giunchiglia, Professor at the University of Trento, and EURAI fellow, member of the Academia Europaea
- Supervision: 1 Master's student on an NLP research project.
- Project: Concept embeddings from lexical knowledge graphs to encode semantics and aligning with transformer model representations; Multilingual lexico-semantic (wordnet-like) resource called UKC, which has 2371 languages, 110K concepts, and 1.9 million words; Data collection mobile application for human-centric AI research;
- Dev Env: Python, PyTorch, Java, NodeJS, PostgreSQL, PostgREST, Mongo, FastAPI(Backend), AngularJS (Frontend), Docker, Kubernetes, Azure

Senior Software Engineer (full-time)

Powered LLC

Jan 2022 – Sep 2023

Ulaanbaatar, Mongolia

- Role: Lead Mobile Application Development (frontend), mobile app architecture solution, infrastructure and planning, research, develop, and adapt new features
- Projects: YESH.MN - Test-based learning for high school students preparing for the university general entrance examination; Octagon Wallet - Mobile application for Octagon Ecosystem - a crypto exchange market, NFT marketplace, and digital wallet; Walk and Earn - Mobile application for the statistics on the walking of the national citizens.
- Supervision: 3 intern developers in mobile application development; client communications, team management, task design, code review, and guiding
- Client: High school students across Mongolia | Octagon Crypto market community
- Dev Env: React NativeJS, ReactJS, GraphQL, Dapp-Blockchain

Software Engineer (full-time)

And Global / Digital Mall LLC

Sep 2019 – Jan 2022

Ulaanbaatar, Mongolia

- Role: Lead Mobile Application Development (frontend), Core backend development (backend) – mobile application architecture solution, infrastructure and planning; Frontend and backend development of the e-commerce platform; Development admin management panel of the e-commerce back-office; Database support engineering
- Dev Env: ReactNativeJS, ReactJS, PHP Symfony, APIplatform, MySQL, AWS

Software Developer (Remote contract)

Alpha Labs LLC

Sep 2023 – Jan 2026

Remote

- Role: Lead Mobile Application Development (frontend)
- Project: 4 different E-commerce mobile applications, widely used in Mongolia, China and Kazakhstan, 1 monitoring app, 1 portal app
- Clients: Delivered these products, all citizens in above mentioned countries.
- Dev Env: React NativeJS, AWS, Google Cloud, Apple, Firebase

EDUCATION

University of Trento

Master of Computer Science; Scholarship: NonEU student fully funded scholarship, 109/110

Trento, Italy

Sep 2023 – Mar 2026

National University of Mongolia

Bachelor of Software Engineering; 3.5/4.0

Ulaanbaatar, Mongolia

Sep 2016 – Jun 2020

TECHNICAL SKILLS

Languages : Python, Java, Javascript, TypeScript, R, C++, PHP, HTML, CSS

Frameworks : PyTorch, Transformers, NLTK, LoRA, Pandas, Scikit-learn, Numpy, React NativeJS, AngularJS, NodeJS, PHP Symfony, FastAPI, PostgREST

Databases : PostgreSQL, MySQL, MSSQL, SparQL, GraphDB, MongoDB
Dev Tools : Git, Docker, AWS, Kubernetes, Firebase, Google Cloud AppStoreConnect, Google Play Console, Test-Flight
Language : English (fluent), Italian (basic)

PROJECTS

<u>Concept Token Embeddings: Research in NLP</u>	<i>Python, PostgreSQL</i>	<i>Sep 2025 - now</i>
* Role: Research and development * Topic: Built concept embeddings from lexical database-knowledge graphs and assess both their capacity to encode semantic information and their correspondence with representations learned by transformer-based language models, and research on methods on validating the method		
<u>Universal Knowledge Core</u>	<i>Python, Java, Spring, PostgreSQL, FastAPI, Docker</i>	http://ukc.datascientia.eu
* Role: DevOps Engineer * Opensource development: building an opensource platform through which community can contribute to the lexical database with new resources * Development: new the importing and exporting lexical resource * Maintenance: sub lexical resource projects for specific languages		
<u>Linguarena</u>	<i>NodeJS, AngularJS, PostgreSQL, Docker</i>	http://linguarena.eu
* Role: Developer and maintenance * Project: Website for searching words from a multilingual dictionary with interactive world map showing cognate words of different languages		
<u>Posture Detection in Sports</u>	<i>Python, MotionCapture</i>	https://github.com
* Role: Data engineer and developer * Analysed the body posture of an archer using MoCap to capture key postural parameters detecting incorrect posture * Results: Provided base method that could lead to an automatic method for evaluating posture and classifying		
<u>Interpreting Sign Language Letters</u>	<i>Java, Processing, PureData, Arduino, Electric circle, Sensors</i>	https://github.com
* Role: Developer * Built a system that recognizes Sign Language alphabet gestures (sensors, electronics and programming)		
<u>YESH.mn - Mobile app</u>	<i>ReactNativeJS</i>	iOS, Android
* Role: Lead Mobile Application development (a project leader) * Results: Used by all high school students preparing for the enrolment examination to universities across Mongolia.		
<u>Octagon Wallet - Mobile app</u>	<i>ReactNativeJS</i>	iOS, Android
* Mobile application for Octagon Ecosystem - a crypto exchange market, NFT marketplace, and digital wallet * Role: Lead Mobile Application development (a project leader)		
<u>Walk and Earn - Mobile app</u>	<i>ReactNativeJS</i>	https://github.com/kbatsuren/wiktra
* Mobile application for the statistics on the walking of the national citizens of Mongolia * Role: Lead Mobile Application development (a project leader)		
<u>BananaMall - E-commerce Platform</u>	<i>PHP symfony, ReactJS, ReactNativeJS, MySQL, APIPlatform</i>	iOS, Android
* Mobile application for the statistics on the walking of the national citizens of Mongolia * Role: Lead Mobile Application development (a project leader), Core backend developer		
<u>iLog</u>	<i>ReactNativeJS</i>	iOS, Android
* Data collection mobile application for human-centric AI research; * Role: Mobile Application developer		
<u>100 ail market - Mobile app</u>	<i>ReactNativeJS</i>	iOS, Android
* E-commerce mobile platform designed to digitize and consolidate Mongolia's building materials trade in Mongolia and China * Role: Lead Mobile Application development (a project leader) * Result: Reduced major traffic congestion in urban areas by supporting the digitalization of marketplace operations, reducing the need for physical visits to trade centers;		
<u>BSB.mn - Mobile app</u>	<i>ReactNativeJS</i>	iOS, Android
* E-commerce mobile application for online shopping and delivery service of BSB group * Role: Lead Mobile Application development (a project leader)		
<u>Taiki - Mobile app</u>	<i>ReactNativeJS</i>	iOS, Android
* Mobile application dedicated to citizens of Kazakhstan for purchasing products from China * Role: Lead Mobile Application development (a project leader)		

- * Results: Established a cross-border e-commerce trade channel between Kazakhstan and China, expanding market access for businesses;

SUVD Online - Mobile app

ReactNativeJS

iOS, Android

- * E-commerce mobile application that imports and resells product from South Korea
- * Role: Lead Mobile Application development (a project leader)

G-Power Monitoring - Mobile app

ReactNativeJS

<https://g-power.mn/>

- * For the clients of a power supply company for monitoring and surveillance of power consumption
- * Role: Mobile Application development

ISU Mongolia - Mobile app

ReactNativeJS

<https://isumongolia.edu.mn/>

- * Mobile portal application for ISU staff and parents to manage school-related activities.
- * Role: Mobile Application development

Driving Exam Test Preparation

ReactNativeJS

Soon to be published

- * Mobile application for preparing the driving theory test exam. Has various functions for preparing the exam and analysing results.
- * Role: Mobile Application development

PUBLICATIONS

- **Yesun-Erdene Jargalsaikhan**, Davide Cavicchini, Fausto Giunchiglia, Jacopo Staiano. "Knowledge-Based Concept Embeddings" (Under review at ACL ARR 2026)
- **Yesun-Erdene Jargalsaikhan**, Davide Cavicchini, Fausto Giunchiglia, Jacopo Staiano. "Semantics over Structure - on the importance of Grounding Concept Embeddings" (Under review at CLiC-it 2026)

CERTIFICATES

- University of Michigan Course Certificate: Introduction to Data Science in Python
- ITPEC - Certified Information Technology Professionals (FE) /MNFE19A0008/ - Fundamental Information Technology Engineer Examination
- ITPEC - Certified Information Technology Professionals (IP) /MNIP18A0009/ - Information Technology Passport Examination

HONOURS AND AWARDS

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| • Employee of the Year, Powered LLC | Mongolia, 12/2022 |
| • And Hackathon 2020, 1st Place : VS Code Extension for Agile Development Support | Mongolia, 10/2020 |
| • Open Platform Hackathon 2019 - LendMN, 2nd Place : Steam Marketplace, Mongolia | Mongolia, 05/2019 |
| • Open Data Hackathon 2019 - National University of Mongolia: Analyses on students' progress | Mongolia, 05/2019 |
| • National Mathematics Olympiad, 3rd Place | Mongolia, 05/2013 |
| • Dev-Summit 2023 (from Developers to Developers), Speaker Topic: Mobile App Development | Mongolia, 08/2023 |
| • National Archery Championship 2023: Compound Archery, <i>Women's Individual</i> 2nd Place | Mongolia, 08/2023 |
| • FISU 2021 World University Games: Compound Archery, <i>Individual & Women team</i> | China, 07/2023 |